Ubaid Bakhtiar

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Ph.D. Candidate, Electrical and Computer Engineering, University of Maryland, College Park

Education

Ph.D in Electrical and Computer Engineering August 2022 – Present
Research Area: Computer Architecture and Domain Specific Accelerators
University of Maryland, College Park Expected Graduation: May 2027
M.Sc in Electrical and Computer Engineering August 2022 – May 2026
University of Maryland, College Park
B.Sc in Electrical Engineering August 2018 – May 2022
Lahore University of Management Sciences (LUMS), Pakistan

Publications

- **Ubaid Bakhtiar**, Amirmahdi Namjoo and Bahar Asgari. **Chasoň: Supporting Cross HBM Channel Data Migration to Enable Efficient Sparse Algebraic Acceleration**
  International Symposium on Microarchitecture (MICRO) 2025 — Acceptance Rate: 20.7%
- **Ubaid Bakhtiar**, Helya Hosseini and Bahar Asgari. **Acamar: A Dynamically Reconfigurable Scientific Computing Accelerator for Robust Convergence and Minimal Resource Underutilization**
  International Symposium on Microarchitecture (MICRO) 2024 — Acceptance Rate: 22.7%
- **Ubaid Bakhtiar**, Jeremy Sha, Helia Hosseini, and Bahar Asgari. **Procyon: Promoting Fine-Grained Multi-Tenancy to Optimize Sparse Streaming Accelerators**
  Design Automation Conference (DAC) 2026 — Acceptance Rate: 20.3%
- **Ubaid Bakhtiar**, Donghyeon Joo and Bahar Asgari. **Pipirima: Predicting Patterns in Sparsity to Accelerate Matrix Algebra**
  Design Automation Conference (DAC) 2025 — Acceptance Rate: 23%
- Helya Hosseini, **Ubaid Bakhtiar**, Donghyeon Joo and Bahar Asgari. **Segin: Synergistically Enabling Fine-Grained Multi-Tenant and Resource Optimized SpMV**
  IEEE Computer Architecture Letters (CAL) 2025 — Acceptance Rate: 20%
- **Ubaid Bakhtiar**, Helya Hosseini, Tom Goldstein and Bahar Asgari. **Orkaria: Offloading Low Rank Adapters for resource-Aware, Memory-Efficient LLM Inference Acceleration**
  Under review
- **Ubaid Bakhtiar**, Sanjali Yadav, Donghyeon Joo, Ramyad Hadidi and Bahar Asgari. **Lyra: Efficiently Supporting Unstructured Sparse LLMs on FPGAs via Variable Matrix Partitioning**
  Under review
- Sanjali Yadav, **Ubaid Bakhtiar**, Helya Hosseini, Donghyeon Joo, Tom Goldstein and Bahar Asgari. **Myla: Maximizing Yield on Lean Hardware for resource-Constrained MoE Inference**
  Under review
- Helya Hosseini, **Ubaid Bakhtiar**, Donghyeon Joo and Bahar Asgari. **Lepus: Leveraging Sparsity to Upend Its Challenges**
  Under review
- Mihir Mahesh, **Ubaid Bakhtiar**, and Bahar Asgari. **Asterope: An Agentic-AI Cycle-Level Simulator for Dataflow-driven Streaming Accelerators**
  Work in progress
- **Ubaid Bakhtiar**, Aninda Manocha, Zhen Jia and Bahar Asgari. **Topology-aware Load Balancer for MoEs on Trainium**
  Work in progress

Work Experience

- **Applied Scientist Intern** – Annapurna Labs, Amazon May 2026 – Present
  - Internship Manager: Dr. Zhen Jia *Cupertino, CA*
  - Working with Neuron Science team at Annapurna Labs, Amazon on optimizing mixture-of-experts (MoE) large

language models (LLMs) on Amazon Trainium 2 and 3 chips. The project involves engaging deeply with the Neuron Kernel Interface (NKI), compiler and communication collectives to maximize hardware utilization.

- **Research Assistant** – Computer Architecture and Systems Lab (CASL), UMD May 2023 – Present
  - Research Area: Computer Architecture and Domain Specific Accelerators College Park, MD
  - Research Advisor: Dr. Bahar Asgari
  - Crafting architectural and system solutions to address computational challenges in sparse and AI workloads, devise methods to improve their performance, and conduct simulation and prototyping on modern architectures such as CPUs, GPUs, and FPGAs.

Professional Services

- Undergraduate Research Mentorship @ University of Maryland, College Park. 2025-2026
- Paper reviewer, ACM Transactions on Computers 2026
- Student Member of Technical Program Committee, IEEE/ACM International Symposium on Microarchitecture (MICRO) 2026
- Student Member of Technical Program Committee, IEEE/ACM International Symposium on Microarchitecture (MICRO) 2026
- Student Volunteer @ SPICE: A Workshop Co-Located with International Symposium on Microarchitecture 2025
- Paper reviewer, IEEE Micro Magazine 2025
- Paper reviewer, ACM Transactions on Reconfigurable Technology and Systems 2025
- Graduate Research Assistant, Computer Architecture & Systems Lab, University of Maryland-College Park
- Graduate Teaching Assistant, Advance Digital Computer Design, University of Maryland-College Park (Fall 2023)
- Graduate Teaching Assistant, Digital Computer Design, University of Maryland-College Park (Fall 2022, Spring 2023)
- Teaching Assistant Feedback Control Systems, LUMS (Spring 2022)
- Teaching Assistant Electromagnetic Fields and Waves, LUMS (Fall 2021)
- Teaching Assistant Engineering Models, LUMS (Fall 2021)
- Teaching Assistant Circuits-II, LUMS (Spring 2021)

Core Competencies

- **Hardware Prototyping:** Hands-on experience in developing custom accelerators for sparse scientific and machine learning workloads. Skilled in CUDA-like execution models, high-level synthesis, placement, and routing on FPGA fabrics, with extensive prototyping of baseline and novel architectures.
- **Performance Modeling:** Development of in-house, cycle-accurate simulators to evaluate architectural tradeoffs in data scheduling, multi-tenancy, and sparsity-aware execution. Experience of performance analysis frameworks such as Xilinx Runtime, CACTI, and Synopsys DC, NVIDIA Nsight Compute and Intel VTune Profiler.
- **Algorithm & Architecture Co-Design:** Identified bottlenecks in sparse stencil solvers and gen AI workloads (e.g., LLMs), bridging algorithmic inefficiencies with architectural design choices to mitigate resource underutilization.
- **Performance optimization.** Expertise in analyzing and optimizing dataflow, memory bandwidth utilization, and compute-resource mapping to close the gap between theoretical peak and achieved performance in sparse accelerators.
- **Tools & Frameworks:** vLLM, Xilinx Vitis, TAPA, Rapidstream, Pytorch, Python, C, C++, Verilog.

Technical Skills

- **Languages:** C, C++, Verilog, Python, OpenCL, MIPS/RISC Assembly, MATLAB
- **Hardware Platforms:** AMD Alveo Accelerator Cards, AMD ZYNQ SoC, RISC, CPU, GPU
- **Frameworks:** vLLM, High-level synthesis, Xilinx Vitis, Xilinx Runtime TAPA, Rapidstream, Intel MKL, Nvidia cuSparse
- **Simulations:** In-house cycle-accurate simulator, CACTI, Synopsys DC, GPGPU-SIM, Nvidia Nsight Compute
- **Deployment:** Linux CLI, Docker, Jupyter, Git
- **Topics:** HW/SW Co-design, Hardware for generative AI, Scientific computing accelerator, AI/ML

Awards and Honors

- **DAC Young Fellow**, Design Automation Conference (DAC) 2026
- **Winner 2-Minute Video Contest**, Student Young Fellow at Design Automation Conference (DAC) 2026
- **Student Travel Grant**, ACM/IEEE 58th International Symposium on Microarchitecture (MICRO 2025)
- **DAC Young Fellow**, Design Automation Conference (DAC) 2025
- **Winner 2-Minute Video Contest**, Student Young Fellow at Design Automation Conference (DAC) 2025
- **Student Travel Grant**, ACM/IEEE 57th International Symposium on Microarchitecture (MICRO 2024)
- **Summer Research Fellowship**, University of Maryland-College Park (Summer 2023)
- **Dean's Honor List Award**, LUMS (2019-20)

Relevant Coursework

- ENEE646 Digital Computer Design (UMD-Fall 2022)
- ENEE645 Compilers and Optimizations (UMD-Spring 2023)
- ENEE640 Digital CMOS Design (VLSI) (UMD-Spring 2023)
- CMSC818J Domain Specific Architecture (UMD-Fall 2023)
- CMSC838L Programming Languages and Computer Architecture (UMD-Spring 2024)
- CMSC722 Artificial Intelligence Planning (UMD-Fall 2024)
- CMSC828G Systems for Machine Learning (UMD-Spring 2025)